Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/18/2025
PI Name:	Stankovic, Aleksandra Ph.D.		
Project Title:	Modeling Individual and Team Problem-Solving Skills for the Establishment of Training Metrics, Standards, and Guidelines for Long-Duration Space Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC018N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	03/09/2020	**End Date:**	08/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/30/2025 per NSSC information (Ed., 12/30/24) NOTE: End date changed to 03/08/2025 per JSC Grant-Technical Officer (Ed., 2/26/24) NOTE: End date changed to 03/08/2024 per A. Beitman/JSC (Ed., 2/6/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robinson, Stephen Ph.D. (University of California, Davis) Strangman, Gary (Massachusetts General Hospital/Harvard Medical School) Caldwell, Barrett (Purdue University)		
Grant/Contract No.:	80NSSC20K0844		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Motivation: Future human lunar exploration will necessitate that astronaut crews work autonomously to solve unexpected and time-critical problems, and will demand increased cooperation between humans and robotic/autonomous systems. While the ability of these multi-agent teams to adapt to unforeseen and off-nominal situations will be integral to mission success and safety, frameworks and recommendations for cultivating team-based problem-solving are currently lacking. Goals: This project aims to characterize team problem-solving strategies for future long duration space exploration tasks, specifically in the context of Lunar and Cis-Lunar operations. This work will generate findings that will help shape team-based concepts of operations and functional requirements, will contribute towards the development of a set of validated measures and metrics for assessing team problem-solving, and will produce an approach to modeling dynamic team problem-solving performance over time, with the goal of helping determine training effectiveness. Outcomes of this study will also help shape future decision and execution support systems. This project has 3 integrated specific aims: Aim 1: Through literature review and in collaboration with teaming subject matter experts, we will (a) identify key teaming and problem-solving knowledge gaps and attributes of interest, and will (b) develop an approach for the in-situ assessment and characterization of individual and team-based problem solving using operationally-deployable metrics and measures. (Completed.) Aim 2: Through data collection in Earth-bound analogs for Lunar/Cis-Lunar activities, we will examine attributes contributing to individual and team problem-solving effectiveness, and will characterize how such skills are developed and used across a variety of operational tasks, under both normal and emergency conditions. (Currently in progress.) Aim 3: Leveraging the insights gained in Aims 1 & 2, we will develop a set of team-based concepts of operations and functional requirements, which may ultimately be used to inform future training guidelines and recommendations for individual and team problem-solving, including for the effective interface with on-board support systems. (Currently in progress.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are undertaking an operationally-relevant assessment of team problem-solving skills in the context of upcoming Cis-Lunar exploration, with the ultimate goal of contributing towards the establishment of future autonomous team-based concepts of operations and functional design requirements for system development, operations, and training.
Task Progress:	Project accomplishments to date include the following items: completion of definition-phase exit milestones (e.g., identification and recruitment of teaming expert co-Investigator, identification of subject matter experts for informational interviewing and data collection, updating of the project timeline and definition of next phase milestones, completion of the Life Sciences Data Archive/LSDA data sharing agreement); completion of preliminary literature review (published Jan 2023 in the journal Human Factors); definition and refinement of experimental methodology and research protocol; definition of field site requirements, identification of potential field sites, and logistical planning for field research, including coordination with NASA Research Operations and Integration (ROI) for the long-duration deployment of our research to an extreme analog site (South Pole Station for winter-over 2023); and drafting of Institutional Review Board (IRB) materials. We have completed data collection with several of our planned extreme operations analog teams, including winter-over crewmembers at the South Pole and a science return team. In collaboration with our field site partners, we are currently collecting additional in-field data with domain-specific expert teams working in the areas of search and rescue/emergency response and remote medical operations. We anticipate completing our field-based data collection across several extreme and remote analog research sites (Aim 2) within the next several months. We are also concluding our distributed team coordination and problem-solving and modeling efforts (Aim 3), and anticipate delivering a computer-based simulation system for team training and dynamic, real-time mission re-planning.
Bibliography Type:	Description: (Last Updated: 03/18/2026)
Abstracts for Journals and Proceedings	Stankovic A, Pollonais J, Thoolen S, Harris K, Davis L, Caldwell B. "Team problem-solving skills for lunar exploration." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025